

1. Smart Document Editor using String, StringBuilder and StringBuffer

The screenshot displays the SIMATS Online Programming Lab interface. On the left, there are two question cards: Q4 (Digital Certificate verification using SET interface, 10 marks) and Q5 (Smart Inventory Management using Map interface, 10 marks). The main area is titled "Your Code" and contains a Java program. The program uses a Scanner to read a paragraph, splits it into words, and then processes each word. It uses StringBuffer to build a new sentence by replacing each word with its length. The output shows the original sentence, the words, and the new sentence where each word is replaced by its length.

```
1 import java.util.*;
2 import java.util.Scanner;
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String paragraph = sc.nextLine();
7         int characterCount = paragraph.length();
8         String[] words = paragraph.split("\\s+");
9         int wordCount = words.length;
10        String[] sentences = paragraph.split("\\.\\?\\!");
11        int sentenceCount = 0;
12        for (String s : sentences) {
13            if (!s.trim().isEmpty()) {
14                sentenceCount++;
15            }
16        }
17        String oldWord = sc.next();
18        String newWord = sc.next();
19        String replaced = paragraph.replaceAll("\\b" + oldWord + "\\b", newWord);
20        StringBuffer history = new StringBuffer();
21        history.append("Replaced ");
22        history.append(oldWord);
23        history.append(" with ");
24        history.append(newWord);
25        history.append("\n");
26        String[] arr = replaced.split("\\.\\?\\!");
27        StringBuilder finalDoc = new StringBuilder();
28        for (String s : arr) {
29            s = s.trim();
30            finalDoc.append(s + ". ");
31        }
32        finalDoc.append("\n");
33    }
34 }
```

Input:

```
Java is fun. I love Java.
Java
python
```

Output:

```
Original Document:
Java is fun. I love Java.
Character : 21
Words : 6
Sentences : 2
```

2. Hospital Token Management System using List Interface

The screenshot displays the SIMATS Online Programming Lab interface. On the left, there are two question cards: Q4 (Digital Certificate verification using SET interface, 10 marks) and Q5 (Smart Inventory Management using Map interface, 10 marks). The main area is titled "Your Code" and contains a Java program. The program uses a Scanner to read patient details (ID, Name, Dept) and stores them in an ArrayList. It then prints all patients and allows for updating a patient's name. The output shows the list of patients and the updated name.

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<Patient> list = new ArrayList<>();
6         int n = sc.nextInt();
7         sc.nextLine();
8         for (int i = 0; i < n; i++) {
9             int id = sc.nextInt();
10            sc.nextLine();
11            String name = sc.nextLine();
12            String dept = sc.nextLine();
13            list.add(new Patient(id, name, dept));
14        }
15        System.out.println("All Patients:");
16        for (Patient p : list) {
17            System.out.println(p.getId() + " " + p.getName() + " " + p.getDept());
18        }
19        int id = sc.nextInt();
20        System.out.println("The Entered patient ID to update is : " + id);
21        sc.nextLine();
22        String newName = sc.nextLine();
23        System.out.println("The Entered name to update is : " + newName);
24        for (Patient p : list) {
25            if (p.getId() == id) {
26                p.setName(newName);
27            }
28        }
29        System.out.println("Updated Patient Details:");
30    }
31 }
```

Input:

```
3
101
Ren
Cardiology
```

Output:

```
All Patients:
101 Ren Cardiology
102 Ren Neurology
103 Ren Cardiology
The Entered patient ID to update is : 101
The Entered name to update is : Ren
```

3. Airport Boarding Queue Management using Queue Interface

SIMATS Online Programming Lab
 JAVA PROGRAMMING

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Q4
 Digital Certificate Verification
 using Set Interface
 4/4 marks

Q5
 Smart Inventory Management
 using Map Interface
 4/4 marks

5/5 answered

Your Code

```

1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         Queue<String> queue = new LinkedList<>();
6         PriorityQueue<String> vip = new PriorityQueue<>();
7         int n = sc.nextInt();
8         sc.nextLine();
9         for (int i = 0; i < n; i++) {
10             queue.add(sc.nextLine());
11         }
12         System.out.println("Current Queue: " + queue);
13         System.out.println("Next Passenger: " + queue.peek());
14         System.out.println("Boarded Passenger: " + queue.poll());
15         System.out.println("Remaining Queue: " + queue);
16         int m = sc.nextInt();
17         sc.nextLine();
18         for (int i = 0; i < m; i++) {
19             vip.add(sc.nextLine());
20         }
21         System.out.println("VIP Queue: " + vip);
22     }
23 }

```

Input
 3
 Ram
 Ravi
 Kumar

Output
 Current Queue: [Ram, Ravi, Kumar]
 Next Passenger: Ram
 Boarded Passenger: Ram
 Remaining Queue: [Ravi, Kumar]
 VIP Queue: [Kumar, Ravi]

Run Save

4. Digital Certificate Verification using Set Interface

SIMATS Online Programming Lab
 JAVA PROGRAMMING

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Q4
 Digital Certificate Verification
 using Set Interface
 4/4 marks

Q5
 Smart Inventory Management
 using Map Interface
 4/4 marks

5/5 answered

Your Code

```

1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<Integer> set = new ArrayList<>();
6         TreeSet<Integer> sorted = new TreeSet<>();
7         int n = sc.nextInt();
8         for (int i = 0; i < n; i++) {
9             int id = sc.nextInt();
10             set.add(id);
11             sorted.add(id);
12         }
13         System.out.println("Certificates (Insertion Order): " + set);
14         System.out.println("Certificates (Sorted Order): " + sorted);
15         System.out.println("Total Unique Certificates: " + set.size());
16         int search = sc.nextInt();
17         System.out.println("The Entered Certificate ID to search is: " + search);
18         if (set.contains(search)) {
19             System.out.println("Certificate Found");
20         } else {
21             System.out.println("Certificate Not Found");
22         }
23     }
24 }

```

Input
 5
 120
 150
 150
 180

Output
 Certificates (Insertion Order):
 [120, 150, 150, 180]
 Certificates (Sorted Order):
 [120, 150, 150, 180]
 Total Unique Certificates: 4

Run Save

5. Smart Inventory Management using Map Interface

12 steps

Q5

Smart Inventory Management
using Map and Map

10 marks

5/5 answered

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         HashMap<Integer, Product> map = new HashMap<>();
6         int n = sc.nextInt();
7         for (int i = 0; i < n; i++) {
8             int id = sc.nextInt();
9             sc.nextLine();
10            String name = sc.nextLine();
11            String category = sc.nextLine();
12            int quantity = sc.nextInt();
13            double price = sc.nextDouble();
14            map.put(id, new Product(id, name, category, quantity, price));
15        }
16        int searchId = sc.nextInt();
17        if (map.containsKey(searchId)) {
18            Product p = map.get(searchId);
19            //System.out.println("Product Found");
20            System.out.println(id + " * " + p.name + " * " + p.category + " * ");
21        } else {
22            System.out.println("Product not Found");
23        }
24        double total = 0;
25        for (Product p : map.values()) {
26            total += p.quantity * p.price;
27        }
28        System.out.println("Total Inventory Value = " + total);
29        sc.nextLine();
30    }
31 }
```

Input:

```
Electronics
5
10000
101
Mouse
```

Output:

```
101 Laptop Electronics 5 10000.0
Total Inventory Value = 50000.0
101 Laptop 5 10000.0
101 Mouse 10 100.0
101 Laptop Electronics 5 10000.0
101 Mouse 10 100.0
```